

Requirements Document

Project Title: AI-Powered Summary Generation for Doctor

Version: 0.1

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1. Introduction

1.1 Purpose

This document serves to point out the functional and non-functional requirements needed for the completion of our AI-powered Summary Generation. These requirements will be the guide for the design, implementation, and testing that will be needed in order to ensure all deliverables meet the expected project objectives.

1.2 Scope

The system allows doctors to upload clinical notes and generate AI-assisted summaries, while redacting confidential information. Doctors can edit summaries and download them as PDF or Word files. The system can be used locally on the user’s computer.

Stakeholders

- Doctors / Scribes (Assistants) (End Users) : needs to summarize patient information.
- (Admin) - Maintains the system itself, as well as the server used for the system itself.
- IT department: maintains connectivity to servers and AI APIs .

Requirements

3.1 Functional Requirements

ID	Requirement	Rationale	Verification Method
FR-1	The system must allow users to submit typed notes.	Using text instead of a document.	Test user text input.
FR-2	The system should allow users to upload multiple files at once.	Multiple summaries required.	Test file input.
FR-3	The system must only store redacted summaries.	Preventing HIPAA non-compliance.	Check the returned summary if it contains redacted information.

FR-4	The system must be able to export the summaries as PDF files or Word documents.	Easier storage of summaries.	Attempt downloading summaries to PDF or Word.
FR-5	The system must allow users to edit summaries after they are generated.	To fix potential summary mistakes.	Attempt to change output summaries.
FR-6	The system should include highlighted references to the original text.	To compare for accuracy and show how the information is linked.	Comparing the original text and the output text.
FR-7	If the AI fails, the system should be able to display the original clinical notes.	To compensate if the system fails.	Test without the AI API connected.

3.2 Non-Functional Requirements

ID	Requirement	Rationale	Verification Method
NFR-1	The system should have a minimal, clean interface.	A simpler interface makes it easier to navigate and use the system.	Test Navigation
NFR-2	The system must generate a summary in 10 seconds or less.	Usability.	Performance Test
NFR-3	The system should have basic authentication methods.	Security.	Login Test
NFR-4	The system should allow users to reset their passwords.	Security.	Login Test
NFR-5	The system should have basic accessibility features as alt text and clear fonts	Usability	Web Accessibility Test
NFR-6	The system should support multiple doctor logins and provide simple role handling. Full access for Doctors, and system setup for Admin.	Manage user access for different system functionalities	Test login for both Doctor and Admin roles, verify that each has appropriate access levels and restrictions.

3.3 Constraints

ID	Constraint	Source
C-1	Accommodate a limited number of users.	Scope Decision
C-2	Cannot be too accurate.	AI Inaccuracies and Time
C-3	Not fully HIPAA-compliant.	Scope Decision
C-4	De-identified or synthetic data will only be used.	Scope Decision and ethical consideration
C-5	Usage of only free/open-source tools.	Budget and resource limitation along with AI Inaccuracies
C-6	Hardware will be limited only to local development laptops.	Resource limitations

Traceability Matrix

Objective	Requirement IDs
Can submit and receive summaries.	FR-1, FR-2, FR-4
Adjustable Summaries with the correct Information.	FR-3, FR-5, FR-6,
Reliable information storage System navigation is cleaner and feedback is faster	NFR-3, NFR-4 NFR-1, NFR-2 NFR-5

Acceptance Criteria

- A doctor is able to create an account, log in, authenticate, and submit text or documents to summarize.
- Translated summaries need to be able to be edited.
- Summaries need to be returned in 10~ seconds.
- Should be able to support multiple Doctors and assistants.
- The summary should achieve an 80% accuracy or above compared to the original document.